UNIT - II : METHODS OF SOIL EXPLORATION

Boring | Auguring | Drilling | Samplers | Soil Sampling

Aim of exploration: (1) know the sequence and thickness of strata, (2) locate the ground water table, (3) collect samples for lab tests, (4) find the depth to bed rock, (5) get strength and compressibility data for design of foundation/earth structures.

1. Stages of a site investigation

- **Reconnaissance / desk study** - maps, aerial photos, geological records, nearby structures.
- **Preliminary exploration** - few boreholes, test pits to get general picture.
- **Detailed exploration** - number, depth and spacing of boreholes planned on the basis of the structure.
- **Testing & report** - field tests (Unit IV), lab tests, final report (Unit V).

Classification of exploration methods

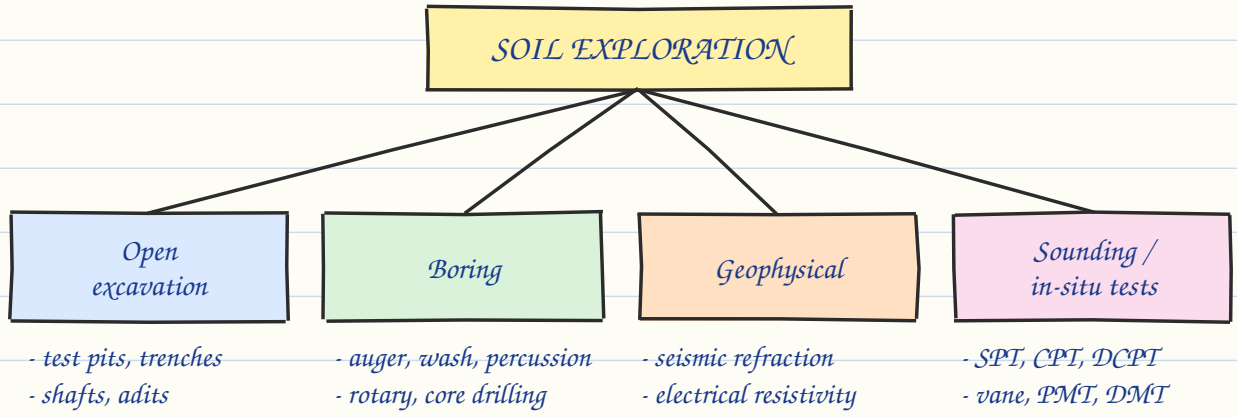


Fig 2.1 Methods of subsurface exploration

1.1 Open excavation methods

- **Test pits (trial pits):** 1.2 m x 1.2 m or more, used to depth 3 - 5 m (dry ground). Soil is seen in-situ; undisturbed block samples can be cut out. Cheap but limited depth and dangerous in loose/wet soil - support the sides.
- **Trenches:** long narrow excavation (1 m wide) to study continuity of strata, fault, pipelines.
- **Shafts:** deep excavation (> 3 m) used for tunnels, large dams and in rock; sides are timbered.
- **Adits / drifts:** horizontal tunnels in hill side - for dam abutments.

1.2 Depth and spacing of boreholes (IS:1892 guideline)

Structure	Spacing of boreholes	Depth of exploration
Isolated building	at least 4 (corners) + 1 at centre	1.5 x width of footing (or 1.5 B) below foundation; to 2 x B for heavy loads
Multi-storeyed buildings	10 - 30 m grid	Till strata of adequate bearing (significant stress < 10 % of qnet)
Highways	30 - 60 m (200 m on uniform ground)	2 - 3 m below subgrade
Earth dams / embankments	30 - 60 m along axis	Up to hard stratum, 2-3 times dam height
Bridges	At each pier/abutment	Up to 2 x B below pier or to rock (min. 3 m in rock)

Rule of thumb: explore to the depth where the increase in vertical stress due to foundation = 10 % (or 20 % in some codes) of the foundation pressure. Continue 3 m into bed rock if rock is met; in case of a floating raft explore below the base up to 1.5 times width.

2. Boring - making a hole in the ground

Boring means advancing a hole (borehole, diameter 50 - 200 mm) to explore the soil in depth. The method chosen depends on soil type, depth, ground water, equipment available and the purpose (sampling / testing / logging only).

Method	How it works	Suitable for
Auger boring	Soil is cut and lifted by auger turned by hand or machine	Soft to stiff cohesive soils above water table; depth up to 6 m (hand) or 30 m (power)
Wash boring	Casing driven; chopping bit on hollow rod, water jetted, cuttings flow up between rod and casing	Sand, silt, soft clays (not gravel/boulders); cheap; sample quality poor
Percussion boring	Heavy chisel / bailer / shell raised and dropped (cable tool)	All soils incl. gravel and soft rock; slow
Rotary drilling	Rotating bit with circulating drilling mud/water/air	Hard soils and rock; deep holes; high speed
Core drilling	Diamond / tungsten carbide bit with core barrel to cut rock cores	Rock; gives cores for RQD, strength tests

2.1 Wash boring (important)

- A casing (50 - 100 mm) is driven by hammer/driving head.
- A hollow drill rod with chopping bit is lowered inside; it is raised and dropped, and rotated by hand to cut the soil.
- Water is forced through the rod by a pump; it comes out of the bit, carries the cuttings up between rod and casing to a settling tank.
- Sample from the tank = disturbed, washed (fines lost). Change of colour/speed is noted to identify strata.
- To take samples, rod is withdrawn and a sampler is attached and driven in.

2.2 Percussion (cable tool) drilling

A heavy tool - chisel (for hard soil/rock), bailer or shell (to remove slurry) - is attached to a wire rope and repeatedly lifted & dropped from a tripod. Water is added to make slurry in dry holes. Good for boulders and gravels but disturbs the soil below the bottom of the hole; used widely for water wells.

2.3 Rotary drilling

- The bit is rotated under axial thrust by a rotary head; drilling fluid (bentonite mud, water or air-foam) is pumped through hollow rods to cool the bit, carry the cuttings and stabilise the borehole wall.
- Bits: drag (fishtail) bit for soft soils, roller (tricone) bit for medium rock, diamond/tungsten carbide bit for hard rock and coring.
- Fast and gives high productivity; hole diameter 50 - 150 mm. Used for depths more than 30 m.
- Core barrels: single tube, double tube (inner tube stays stationary, protects core), triple tube; sizes EX, AX, BX, NX (NX core dia = 54 mm).

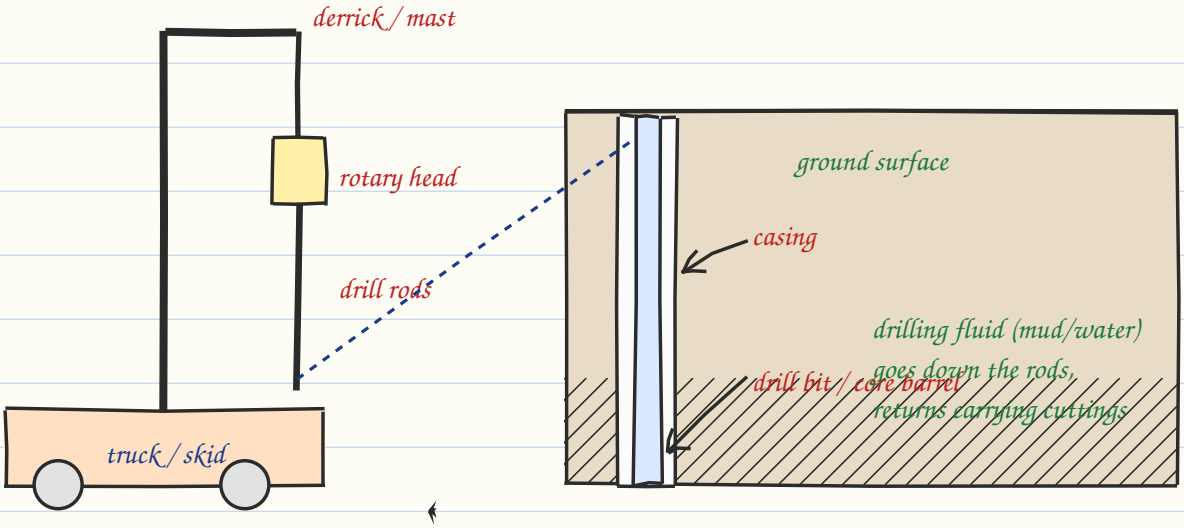


Fig 2.2 Rotary drilling rig with circulating fluid

3. Auguring

Auger boring is the simplest method; the auger is rotated into ground and soil is lifted on its flights. It is dry boring - used above ground water. In loose sand below water table the hole collapses, so casing is used.

3.1 Types of augers and their usage

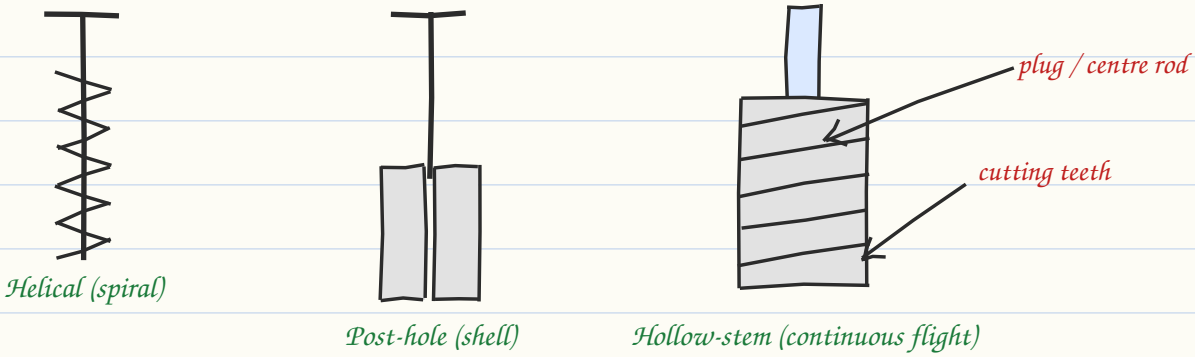


Fig 2.3 Common types of augers

Type	Description	Use
Helical (spiral) auger	Screw-like, 75 - 200 mm dia, hand operated with T-handle, extensions of 1 m	Stiff, dry, cohesive soil up to 6 - 10 m; shallow investigation, road projects
Post-hole (shell) auger	Two half-cylinders with cutting edges, like a bucket	Soft to medium clay, silt above W.T.; faster, brings up sample
Iwan (Dutch) auger	Open-ended cylindrical, 100 - 200 mm; hand rotated	Soft and sandy soils, shallow
Continuous flight (power) auger	Machine rotated long flights, soil comes up the flights continuously	Quick logging to 30 m; highways, shallow foundations
Hollow-stem auger	Flights wound on a hollow tube; plug removed to insert sampler/SPT through it	Soils with water table, caving sands - gives clean access for SPT & thin-wall sampling
Bucket (large dia) auger	Up to 1 m diameter bucket-like	Bored pile excavation, inspection of strata by person lowered

Advantages of auger boring: cheap, rapid, small equipment, no water needed.
Limitations: unsuitable below W.T. in sands, gravels or boulders; samples are disturbed and mixed; depth of strata change is only approximate.

Machinery used for drilling

- Hand operated: tripod, pulley, hand augers, hammers - small jobs, up to 10 m.
- Truck / skid mounted rotary rigs: engine + hydraulic feed + rotary table (top-drive), mast, mud pump, winch.
- Percussion (churn) rigs: tripod/derrick with winch & spudding beam.
- Wire-line drill for deep core drilling, sonic drills for continuous cores in difficult ground, dual-tube reverse circulation for gravel & boulders.
- Accessories: casing, drill rods (AW, BW), bits, shoes, mud pump, water tank, tripod & pulley, SPT hammer (63.5 kg), bailers, tremie, core boxes.

4. Soil sampling

The quality of design depends on the quality of samples. A sample must represent the soil as found in the field. Disturbance may be caused by the drilling method, sampler wall thickness, stress relief, handling and storage.

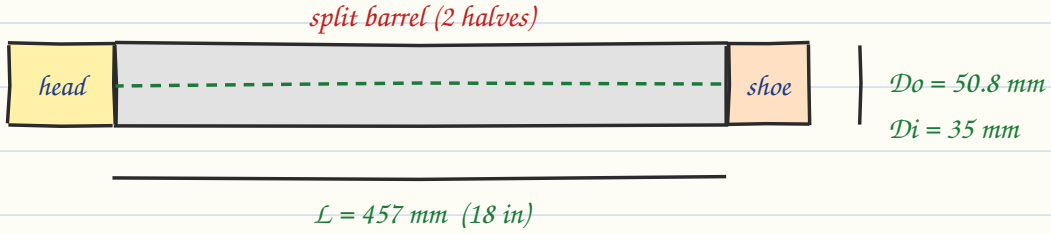
4.1 Types of samples

Sample	Description	Used for
Disturbed (non-representative)	Structure destroyed; composition may change (fines lost); collected from cuttings, auger, wash water	Visual identification only (logging)
Disturbed (representative)	Structure destroyed but all constituents and water content preserved (split spoon, auger cuttings sealed in jar)	Grain size, Atterberg limits, specific gravity, water content, compaction
Undisturbed	Natural structure, density & water content are preserved as far as practical (thin-wall tube, block, piston sample)	Shear strength, consolidation, permeability, unit weight
Block samples	Cut by hand from test pit or trench, waxed immediately	Best quality for clay, residual soils, sensitive/stiff soils
Core samples	Cylinder of rock from core barrel	RQD, compressive strength, rock tests

Truth about undisturbed samples: a truly undisturbed sample is impossible - stress relief, water change and sampling stresses always exist. So we say "undisturbed samples are those with minimum disturbance" (Hvorslev). Rule: design-quality sample = area ratio < 10 %, inside clearance 1 - 3 %, recovery ratio > 95 %.

4.2 Sampling methods / samplers

IS:2131 / ASTM D1586 standard split spoon

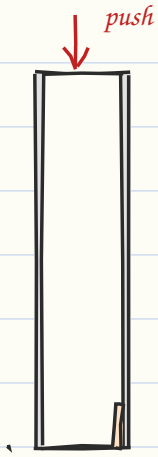


Ends in cutting shoe; ball valve in head vents water; sample retained by core-catcher.

Fig 2.4 Standard split-spoon sampler

- *Split-spoon (split barrel) sampler:* 50.8 mm OD, 34.9 mm ID, 457 mm long barrel; heavy walls -> disturbed but representative sample. Also used in SPT. Liner may be used.
- *Thin-walled (Shelby) tube sampler:* seamless steel tube, 50 - 100 mm OD, wall 1.25 - 3 mm, *pushed* (not hammered) steadily into soft to medium clays. Ends sealed with wax after taking out.
- *Piston sampler (stationary piston / Osterberg / Hvorslev):* thin tube with piston which keeps fluid pressure off the sample when it is pushed; for soft sensitive clays, recovery near 100 %.
- *Denison sampler:* double-tube rotating core barrel with inner non-rotating liner; for stiff clays and hard soils.
- *Pitcher sampler:* spring-loaded inner tube projects ahead of the outer rotating tube; for stiff to hard clays, and soft rock.
- *Open drive sampler:* simple open tube driven by hammer; *Foil / Swedish sampler:* for very soft clays - continuous samples.

- Scraper bucket sampler: for sampling gravelly, sandy soil below water.
- Rotary core barrels (NX etc.) with diamond bits: rocks.



Shelby tube (thin walled)

wall 1.25 - 3 mm steel / brass

length = 5 to 10 x diameter

Di = inner dia at cutting edge, De = Dc = outer dia

cutting edge is sharp (taper 6 - 10 deg)

Common sizes: 50 mm, 75 mm, 100 mm OD

Pushed (not driven) into soft/medium clay

Fig 2.5 Thin walled sampler

4.3 Sampler specifications - design parameters

Disturbance depends on the geometry of cutting edge and wall, defined by the following ratios (where Do = outer dia of tube at cutting edge, Di = inner diameter at cutting edge, Dc = inside dia of sampler tube above cutting edge, De = outside dia of tube):

Parameter	Formula	Permissible for undisturbed sampler
Area ratio Ar	(De^2 - Di^2)/Di^2 x 100	< 10 % (soft clay) ; 10-20 % (stiff clay / sand); split spoon about 100 %
Inside clearance Ci	(Dc - Di)/Di x 100	0.5 - 3 % (to reduce friction on the sample)
Outside clearance Co	(Do - De)/De x 100	0 - 2 % (to reduce driving effort)
Recovery ratio Lr	Length of sample recovered/length of tube pushed	> 0.96 for good sample
Length/diameter	L / D	Clay: 5 - 10 ; sand: up to 20

Example: Thin tube with De = 54 mm, Di = 50 mm: Ar = (54^2 - 50^2)/50^2 x 100 = (2916 - 2500)/2500 x 100 = 16.6 % (> 10 %, so only for stiff clay, moderate quality). For a split spoon De = 50.8, Di = 34.9: Ar = (2580.6 - 1218)/1218 = 112 % -> highly disturbed.

4.4 Sample sizes (usual)

Test	Minimum sample size
Visual / identification	500 g from each stratum (jar)
Grain size, Atterberg limits, G	1 - 2 kg disturbed soil
Compaction (Proctor)	25 - 30 kg bulk sample
Unconfined / triaxial test	38 mm dia x 76 mm length (min. 3 specimens) from 75 mm tube
Consolidation test	60 mm dia x 20 mm specimen (from 75 - 100 mm tube)
Rock compressive strength	NX cores, L/D = 2 : 1
Water sample	1 litre in clean bottle (chemical tests)

5. Handling, preservation, storage and transport of samples

- Label every sample immediately: project, borehole no., depth, date, sampler, sample no., top & bottom, method, water table. Keep a separate record.
- Disturbed samples: place in air-tight jars / polythene bags; add label inside and outside; protect from sun to keep natural moisture.
- Undisturbed samples: seal both ends of the tube with molten wax (3 layers, 25 mm thick) + cap; store vertically, top up. For block samples: wrap in cloth, dip in wax, box with sawdust.
- Storage: cool, shaded, humidity-controlled room; avoid freezing/vibration. Test within 2 - 3 weeks (soft clays within 1 week) to avoid water migration & ageing.

- Transport: use padded wooden boxes, cushion of foam/sawdust, handle with care - NO jolt or dropping; keep tubes vertical.
- Extrusion in the lab should be in the same direction as the sample entered the tube.
- Retain all rock cores in core boxes with depth marking; photograph them.

5.1 Sample preparation in laboratory

- Disturbed: air-dry (or oven-dry below 60 deg C for organic soils), break lumps with wooden mallet, pass through 4.75 mm sieve, use quartering (coning) to get representative portion.
- Undisturbed: extrude, cut specimen with wire saw, trim in lathe to required dia and length, keep covered with wet cloth to avoid drying.
- Moisture content of each specimen should be taken from the trimmings.

5.2 Factors causing disturbance of samples

- Method of boring: driving casing, jetting water below the casing tip, over-drilling or heave in the hole bottom (sand "boiling").
- Stress relief: removal of overburden pressure -> soil swells, structure loosens.
- Sampler design: thick wall, large area ratio, blunt cutting edge, no inside clearance -> shear distortion of edges of sample.
- Sampling operation: hammering instead of pushing, jerky push, leaving tube in hole (suction on withdrawal) - use slow steady push and rotate before pulling.
- Handling: vibration in transport, sun heat, loss of moisture, extrusion in wrong direction, long storage.

5.3 Quality classes of samples (Hvorslev)

Class	Quality	Typical properties obtainable
1	Undisturbed (block, piston)	Strength, stiffness, consolidation, permeability
2	Slightly disturbed (thin tube)	Water content, unit weight, strength (approx.)

Class	Quality	Typical properties obtainable
3	Remoulded - all constituents retained (split spoon)	Grading, Atterberg limits, water content, G
4	Mixed / washed fines lost	Only soil type, rough grading
5	Debris / cuttings	Strata boundary identification only

5.4 Choice of drilling method - comparison

Method	Merits	Demerits
Auger	Cheap, quick, simple	Limited depth; fails below W.T. in sand/gravel
Wash boring	Low cost, hand rig, soft soils to 30 m	Sample disturbed; unsuitable for gravel/boulders; slow
Percussion	Penetrates boulders & hard layers	Slow, disturbs soil below bit, no good sample
Rotary	Fast, deep, hard formations, core	Costly, heavy equipment, mud disposal

5.5 Stabilising the borehole

- Casing: steel pipe driven or placed to prevent caving; withdrawn after finishing (or left for piezometer).
- Drilling mud (bentonite slurry): forms a thin mud cake on the wall, hydrostatic pressure holds the wall open - no casing required in many soils; density ~ 1.05 - 1.2 t/m3.
- Keep water level in casing higher than ground water table to avoid quick condition / piping and loosening of sand at the bottom (very important before SPT).
- Backfill all exploratory holes with cement-bentonite grout / cuttings after use to protect ground water & safety.

5.6 Field safety and good practice

- Locate underground utilities before boring (power cables, gas, water lines).
- Record rig down-time, drilling rate, losses of drilling fluid (sign of cavities or fractured rock).

- Mark borehole coordinates with GPS/ total station and reduce level (R.L.) to a benchmark.

6. Summary table

Soil	Boring	Sampling
Soft clay	Auger / wash boring	Thin-wall tube / piston sampler
Stiff clay	Rotary / auger	Denison, Pitcher, block sample
Sand (dry)	Auger / hollow stem	Split-spoon (SPT) - disturbed
Sand below W.T.	Wash / rotary with mud	Split-spoon, piston (specialised), freezing
Gravel, boulders	Percussion	Bulk sample, scraper bucket
Rock	Rotary core drilling	Diamond core barrel (NX)

7. Important questions

- Describe various methods of boring. Explain wash boring with a neat sketch.
- Write short notes on types of augers and their uses.
- What is an undisturbed sample? Explain how disturbance is reduced in a sampler (area ratio, clearances).
- Explain the procedure of handling, storing and transporting samples.
- A thin-wall sampler has De = 76 mm, Di = 72 mm. Compute area ratio and comment. (Ar = 11.4 %)
- Differentiate split-spoon and Shelby tube sampler.